

Qatar-1b

R-1643

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250505 · created 2026-08-02T07:07:37+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460800.8964 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.017
Transit depth (Rp/Rs)^2	3.64 +/- 0.65 [%]
Orbital Inclination (inc)	82.64 +/- 1.56
Airmass coefficient 1 (a1)	0.951 +/- 0.019
Airmass coefficient 2 (a2)	0.036 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.02 %
Best Comparison Star	#3 - [364, 197]
Optimal Aperture	3.77
Optimal Annulus	7.79
Transit Duration (day)	0.062 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.017 vs 0.1463 (deviation 2.3 σ)
Duration fit vs published	0.062 d vs 0.0692 d (deviation 0.45 σ)
Mid-time O-C	-7.7 min
Depth SNR	9.8
Residual scatter	2.02 %

Light curve

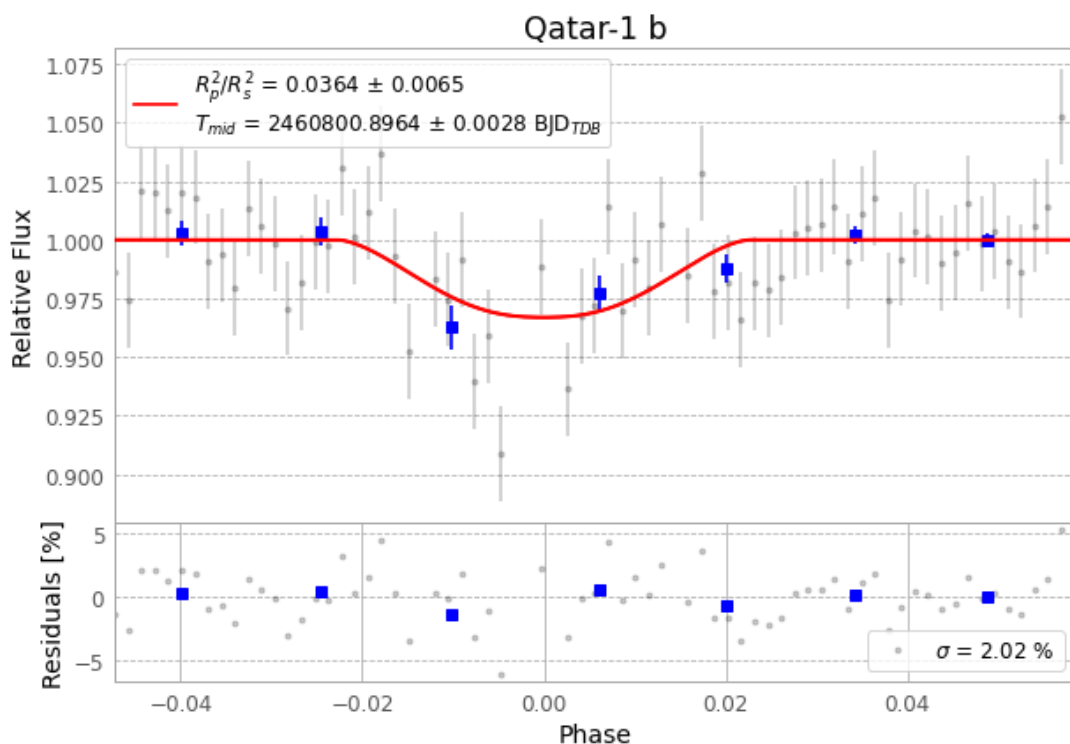


Figure 1 — FinalLightCurve_Qatar-1 b_2025-05-05.png (SHA-256 cde3865f47c94a98...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8b7d49ba88dac4ab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), Qatar-1250505075416.FITS → Qatar-1250505113015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250505.log (ad2ab83d9a38...), FinalLightCurve_Qatar-1 b_2025-05-05.png (cde3865f47c9...), FinalLightCurve_Qatar-1 b_2025-05-05.csv (d7a7fe34910a...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 07:07Z.